Python Programming

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1. Explain the history and evolution of Python. Discuss its design philosophy, key versions, and how it has become one of the most popular programming languages today.
2. What is an Integrated Development Environment (IDE)? Compare and contrast at least two popular Python IDEs, highlighting their features and suitability for different types of projects.
3. Python is known for its clean and readable syntax. Explain the importance of **indentation** in Python with examples. How does it differ from other languages that use braces?
4. Define **variables** in Python. Explain different **data types** available (e.g., numeric, sequence, mapping, boolean) with examples. How does Python handle dynamic typing?
5. Write a Python program to demonstrate **string manipulation** Include operations like slicing, concatenation, repetition, membership testing, and using common string methods (e.g., `upper()`, `split()`, `replace()`).
6. Explain implicit and Explicit Type-casting
7. What are local and global scopes in Python? Illustrate with examples how variable scope works inside and outside functions. What is the `global` keyword and when is it used?
8. Explain conditional statements in Python (`if`, `elif`, `else`). Write a program to categorize a person's age into child, teenager, adult, or senior citizen.
9. Describe loops in Python (`for` and `while`). Compare their use cases. Write a program to print the multiplication table of a given number using both types of loops.
10. What are jump statements in Python? Explain `break`, `continue`, and `pass` with suitable examples.
11. Explain the `match-case` statement introduced in Python 3.10. How is it different from traditional `if-elif-else` chains? Write a program for a simple calculator using `match-case`.
12. Define functions in Python. Explain the difference between defining a function and calling it. Discuss parameters, arguments, and the use of the `return` keyword .
13. Write a function-based program to check if a number is prime. Then, extend it to print all prime numbers up to `n`. Explain the algorithm used.

14. Create a pizza order program that takes user input for size, toppings, and extra cheese, calculates the total bill based on a price chart, and displays an itemized receipt. Use functions to modularize the code.
15. Write a program to check if a given string is a palindrome . Ignore case, spaces, and punctuation. Explain your approach step by step.
16. Python is used in various real-world applications . Discuss at least three major domains (e.g., web development, data science, automation) where Python is extensively used, giving examples of popular libraries/tools in each domain.
17. How does Python's memory management work for different data types? Explain with examples the concept of mutable vs. immutable objects and how it affects variable assignment and function arguments.
18. Design a menu-driven program using functions for a library system with options to:
    - [ ] Add a book (title, author).
    - [ ] Display all books.
    - [ ] Search for a book by title.
    - [ ] Exit.

    Use loops, lists, and string operations.